WORKSHEET 9.7

Free-Response Type: Differential Equations

Unit 9: Exam Strategy and Free-Response Types

Key ideas

- A slope-field segment at (x, y) has slope equal to the value of $\dfrac{dy}{dx}$ there.
- A tangent-line approximation uses the given point and the slope from the equation.
- $\dfrac{d^2y}{dx^2}$: differentiate the equation implicitly, then substitute for $\dfrac{dy}{dx}$. Its sign decides over or under.
- To solve: separate, integrate with $+C$, use the initial condition, then solve for y. No separation, no points.

Worked examples

Example 1 · A tangent-line approximation

Let $y = f(x)$ be the solution of $\dfrac{dy}{dx} = 2x(y-1)$ with $f(1) = 4$. Approximate $f(1.1)$.

Solution. Slope at $(1, 4)$: $2(1)(3) = 6$. Tangent line: $y = 4 + 6(x-1)$.

$f(1.1) \approx 4.6$.

Example 2 · Over or under

Is that approximation an overestimate or an underestimate?

Solution. $\dfrac{d^2y}{dx^2} = 2(y-1) + 2x\dfrac{dy}{dx}$, which is $6 + 12 = 18$ at $(1, 4)$.

An underestimate: the graph is concave up near $x = 1$.

Example 3 · The particular solution

Find $f(x)$.

Solution. $\dfrac{dy}{y-1} = 2x\,dx$ gives $\ln|y-1| = x^2 + C$, and $\ln 3 = 1 + C$.

$y = 1 + 3e^{x^2-1}$.

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Practice

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CORE

1. For $\dfrac{dy}{dx} = x - 2y$, find the slope at $(3, 1)$.

2. For the same equation, along what line are the slope-field segments horizontal?

3. For the same equation, find $\dfrac{d^2y}{dx^2}$ in terms of x and y.

4. Show that $y = \dfrac{x}{2} - \dfrac{1}{4}$ is a solution of $\dfrac{dy}{dx} = x - 2y$.

EXAM LEVEL

5. Solve $\dfrac{dy}{dx} = \dfrac{2x}{e^y}$ with $y(0) = 0$.

6. Solve $\dfrac{dy}{dx} = -xy^2$ with $y(0) = 2$.

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7. Solve $\dfrac{dP}{dt} = 0.5(80 - P)$ with $P(0) = 20$, and find $\lim\limits_{t \to \infty} P(t)$.

8. Find the values of m and b for which $y = mx + b$ is a solution of $\dfrac{dy}{dx} = y - 4x$.

Question 9 · Free-response practice

Consider the differential equation $\dfrac{dy}{dx} = x(4 - y)$. Let $y = f(x)$ be the particular solution with $f(2) = 1$.

(a) Write an equation of the line tangent to the graph of f at $(2, 1)$, and use it to approximate $f(2.1)$.

(b) Find $\dfrac{d^2y}{dx^2}$ in terms of x and y. Is the approximation in part (a) an overestimate or an underestimate? Give a reason.

(c) Find the particular solution $y = f(x)$.

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Worked solutions

1. $3 - 2 = 1$.
2. $y = \dfrac{x}{2}$.
3. $1 - 2\dfrac{dy}{dx} = 1 - 2x + 4y$.
4. $\dfrac{dy}{dx} = \dfrac{1}{2}$, and $x - 2y = x - x + \dfrac{1}{2} = \dfrac{1}{2}$.
5. $e^y\,dy = 2x\,dx$: $e^y = x^2 + C$ with $C = 1$, so $y = \ln(x^2 + 1)$.
6. $-y^{-2}\,dy = x\,dx$: $\dfrac{1}{y} = \dfrac{x^2}{2} + C$ with $C = \dfrac{1}{2}$, so $y = \dfrac{2}{x^2 + 1}$.
7. $-\ln|80 - P| = 0.5t + C$; $80 - P = 60e^{-0.5t}$, so $P = 80 - 60e^{-0.5t}$ and the limit is 80.
8. $m = mx + b - 4x$ for all x requires $m = 4$ and $b = 4$: $y = 4x + 4$.
9. (a) Slope $2(4 - 1) = 6$: $y = 1 + 6(x - 2)$, so $f(2.1) \approx 1.6$.

   (b) $\dfrac{d^2y}{dx^2} = (4 - y) - x\dfrac{dy}{dx} = (4 - y)\left(1 - x^2\right)$, which is -9 at $(2, 1)$. The graph is concave down there, so the approximation is an overestimate.

   (c) $\dfrac{dy}{4 - y} = x\,dx$ gives $-\ln|4 - y| = \dfrac{x^2}{2} + C$, and $-\ln 3 = 2 + C$. So $4 - y = 3e^{2 - x^2/2}$ and $y = 4 - 3e^{2 - x^2/2}$.

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